Dermatoses Due to Plants 17

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Key features

- Plant constituents cause a variety of adverse reactions, including many cases of allergic contact dermatitis
- Plant dermatoses are due to a small minority of plants
- Urticaria, mechanical and chemical irritant contact dermatitis, phytophotodermatitis, and allergic contact dermatitis are the most common plant dermatoses
- Two major plant allergens are urushiol (Anacardiaceae family which includes poison ivy) and sesquiterpene lactones (Asteraceae family which includes chrysanthemums)

Important botanical synonyms

New family name	Old family name
Apiaceae	Umbelliferae (Parsnip family)
Arecaceae	Palmae (Palm family)
Asteraceae	Compositae (Daisy family)
Brassicaceae	Cruciferae (Mustard family)
Fabaceae	Leguminosae (Pea family)
Lamiaceae	Labiatae (Mint family)
Poaceae	Gramineae (Grass family)

Rhus ≠ *Toxicodendron* or poison ivy (see text)

Non-immunologic urticaria = Toxin-mediated urticaria

INTRODUCTION

Contrary to the belief of patients, just because a product is "natural" and not "artificial", it doesn't mean it is safe. When patients discover that a "natural" skin care product is harming them, they are dumbfounded. The light only goes on in their minds when we point out that, "Poison ivy is natural, too." All the reactions described in this chapter are natural.

Plant reactions can be classified into urticaria (immunologic and toxin-mediated; see Chs. 16 & 18), irritant dermatitis (mechanical and chemical), phototoxic dermatitis (phytophotodermatitis), and allergic contact dermatitis. Table 17.1 lists the most common plant reactions.

Botanical Aspects

Becoming familiar with basic botanical nomenclature used for the over 390 000 plant species will help clinicians to better understand the literature that addresses plant-related dermatoses[1].

Poison ivy – just as allergenic despite a name change

It is occasionally necessary to change a plant's name. For example, the genus *Rhus* was once thought to be a cohesive, widespread genus that included poison ivy (previously known as *Rhus radicans* or *Rhus toxicodendron*). Research in the 1950s and 1960s revealed that poison ivy and its immediate relatives represented a distinct developmental lineage that should not be classified as *Rhus* but as a distinct genus, *Toxicodendron*, and thus the correct scientific name for common poison ivy is *Toxicodendron radicans*.

Binomial nomenclature

The nomenclatural scheme of botany and zoology gives every entity two names: a generic name followed by a specific epithet (e.g. *Toxicodendron radicans*). A reference to the person (authority) who first published the species name is often placed after the binomial name. The full binomial name of common poison ivy reads "*Toxicodendron radicans* (L.) O. Ktze". "(L.)" refers to Carl Linnaeus, who first named the plant with the specific epithet *radicans*. "O. Ktze" refers to Otto Kuntze, who transferred the epithet *radicans* from the genus *Rhus* into the genus *Toxicodendron*.

The genus, species, and any infraspecific epithets are called the "minor categories of nomenclature", and these names are used when citing a plant's name (Table 17.2). Genera are grouped into families, and if you know the family to which a plant belongs, you will probably be able to determine what kind of dermatologic manifestations (if any) it can cause.

Plant identification

The following recommendations should be followed to identify an unknown plant that you or your patient thinks is causing an eruption.

- *Collect specimens*. Herbs (low-growing plants with fleshy stems) should be collected in their entirety with their flowering and/or fruiting materials intact. For shrubs (low-growing plants with woody stems), the end of a branch (up to 60 cm) should be collected, with the leaves attached, as well as any flowering or fruiting materials present. You may press plants before submitting them for identification, or you can put them into a plastic bag with *no water added* and submit them with data on: (1) where that plant was collected; and (2) its habitat (cultivated in a garden; growing wild along a creek bank, in an open field, in high-elevation woodlands, etc.).
- *Involve a plant taxonomist*. Every US state and most Canadian provinces have an established mechanism for routine and rapid plant identification, and in most locales the service is free. In suburban and rural areas, the county or regional agricultural extension service or farm advisors are knowledgeable about the procedures for immediate plant identification, and they have access to the regional herbaria. Urban areas have similar services, usually accessed through the urban horticulturist of the state extension service, or through the local garden center or botanical garden. In other countries, look for a nearby herbarium housing a collection of flora of the region. A botanist who works there would likely be willing to help identify an unknown plant.

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MOST COMMON PLANT DERMATOSES			
Binomial name	**Common name**	**Reaction type**	**Inciting agent**
Allium sativum	Garlic	ACD	Diallyl disulfide, allicin
		ICD	Probably the same
Alstroemeria spp.	Peruvian lily	ACD	Tulipalin A > B
Capsicum annuum	Hot peppers	Erythema/edema/burning	Capsaicin
Citrus latifolia	Persian lime	Phytophotodermatitis	Furocoumarins
Narcissus pseudonarcissus	Daffodils	ICD	Calcium oxalate
Opuntia spp.	Prickly pear and others	Mechanical irritant dermatitis	Glochids
Parthenium hysterophorus	Scourge of India, congress grass	ACD	Sesquiterpene lactones
Toxicodendron radicans	Poison ivy	ACD ICD (black-spot reaction)	Alkyl-catechols and resorcinols in urushiol
Toxicodendron diversilobum	Poison oak	ACD ICD (black-spot reaction)	Alkyl-catechols and resorcinols in urushiol
Tulipa spp.	Tulips	ACD	Tulipalin A > B
		ICD	Coarse hair on bulbs
Urtica dioica	Stinging nettle	Urticaria	Histamine
X *Dendranthema* cultivars	Chrysanthemums	ACD	Sesquiterpene lactones

Table 17.1 Most common plant dermatoses. These are admittedly biased by the anecdotal and unscientific observations as well as literature reviews of the author. ACD, allergic contact dermatitis; ICD, irritant contact dermatitis.

THE BASIC NOMENCLATURAL SCHEME FOR PLANTS, USING COMMON POISON IVY AS THE EXAMPLE	
Kingdom	**Plantae (Metaphyta)**
Division	Magnoli<u>ophyta</u> (flowering plants)
Class	Magnoli<u>opsida</u> (dicotyledons)
Subclass	Rosi<u>dae</u>
Order	Sapind<u>ales</u>
Family	Anacardi<u>aceae</u> (sumac or cashew family)
Genus	*Toxicodendron*
Species	*radicans*
(Author's name)	(L.) O. Ktze

Table 17.2 The basic nomenclatural scheme for plants, using common poison ivy as the example. The distinctive ending for each category is underlined. Note that the minor categories of nomenclature have no special endings.

Patch Testing

Details of patch testing are beyond the scope of this chapter (see Ch. 14). The reader is also referred to the paper by Mitchell[2]. The most common allergenic plants and a proposed botanical screening tray are found in Table 17.3[3].

MOST COMMON RELEVANT BOTANICAL ALLERGENS AND PROPOSED BOTANICAL SCREENING TRAY		
Plant name	**Plant family**	**Test compound**
Achillea millefolium (yarrow) *Arnica* spp. *Chrysanthemum* cultivars *Matricaria chamomilla* (chamomile) *Tanacetum parthenium* (feverfew)	Asteraceae	Compositae mix II 6% in petrolatum, sesquiterpene lactones (SQLs) mix 0.1% in petrolatum, and parthenolide 0.1% in petrolatum
Alstroemeria spp.	Alstroemeriaceae	Alpha-methylene-gamma-butyrolactone 0.1% in petrolatum
Cananga odorata	Anonaceae	Cananga (ylang-ylang) oil 2% in petrolatum
Citrus aurantium var. *amara*	Rutaceae	Neroli oil 4% in petrolatum
Lavandula spp.	Lamiaceae	Lavender (absolute) 2% in petrolatum
Lichens (fungus–alga dual organisms)	Non-plant	Lichen acid mix 0.3% in petrolatum (contains usnic acid, atranorin, evernic acid)
Melaleuca alternifolia	Myrtaceae	Tea tree oil 5% in petrolatum
Mentha spp.	Lamiaceae	Spearmint oil 5% in petrolatum
Pelargonium cultivars	Geraniaceae	Geranium oil (Bourbon) 2% in petrolatum
Primula obconica	Primulaceae	2-methoxy-6-n-pentyl-4-benzoquinone (primin) 0.1% in petrolatum
Rosa damascena	Rosaceae	Rose oil (Bulgarian) 2% in petrolatum
Santalum album	Santalaceae	Sandalwood oil 2% in petrolatum
Taraxacum officinale	Asteraceae	Dandelion 2.5% in petrolatum

Table 17.3 Most common relevant botanical allergens and proposed botanical screening tray. This list may help to narrow down a potential botanical allergy in a patient who brings specimens into your office to examine[3]. Note this screening tray does not include urushiol. See Appendix in Chapter 14 for commercial sources of patch tests.

IMMUNOLOGIC CONTACT URTICARIA

Key features

- Atopy and frequent contact with fresh fruits and vegetables are risk factors
- May present as urticaria, pruritus, burning, or chronic dermatitis
- Oral allergy syndrome is mucosal contact urticaria caused by antigens similar to allergenic pollen
- Protein contact dermatitis represents an eczematous eruption arising from repeated urticarial reactions

Plant-induced urticarial reactions are divided into immune and non-immune (toxin-mediated [see next section]). Several cutaneous reaction patterns besides wheals are possible, e.g. erythema, dermatitis.

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Epidemiology

Immunologic contact urticaria is rather uncommon. Approximately 95% of cases are work-related – long-time food handlers with underlying dermatitis are at greatest risk. However, fully half of patients with "protein contact dermatitis", a type IV eczematous eruption arising from repeated type I urticaria reactions, are not atopic.

Examples of reported urticants include common vegetables (e.g. celery, onions, potatoes, lettuce), fruits (e.g. tomatoes, bananas, lemons), herbs (e.g. parsley, dill) and nuts, as well as algae, lichens, shrubs, trees, and grasses.

Pathogenesis (see Ch. 18)

IgE-mediated release of vasoactive mediators from mast cells leads to local urticaria and, rarely, a "contact urticaria syndrome" that includes local wheals plus systemic symptoms involving the nose, throat, lungs, gastrointestinal tract, or cardiovascular system. The main cause seems to be histamine release, but prostaglandins, kinins, and leukotrienes probably augment the inflammatory response.

Clinical Features

Within 30 minutes of contact with certain fresh foods, affected individuals experience pruritus, erythema, urticarial swelling, and even dyshidrotic-like vesicles. Sometimes, individuals only develop burning or tingling without objective findings[4]. Theoretically, any plant can cause contact urticaria, especially with repeated exposures on the wet, macerated skin of food handlers. Cooking, processing, deep-freezing, or crushing fruits and vegetables generally reduces their allergenicity.

Some individuals become cross-sensitized to pollen and similar allergens in fruits or vegetables[5]. Upon eating a cross-reacting food, they experience sudden, IgE-mediated, oral cavity itching, stinging, and pain. Edema of the lips, tongue, palate, and pharynx typically ensue as the "oral allergy syndrome" (OAS) progresses. Gastrointestinal symptoms and anaphylaxis are possible if enough allergens are ingested. For example, 70% of European patients with immediate hypersensitivity to birch pollen develop OAS while eating apples, pears, cherries, peaches, plums, apricots, almonds, celery, carrots, potatoes, kiwis, hazelnuts, or mangoes. Pollen-associated foods are often, but not always, edible when heated.

The term "protein contact dermatitis" describes a chronic dermatitis in which patch tests are typically negative but prick tests to large protein allergens are positive[6]. Patients develop a chronic dermatitis – frequently on the hands and fingertips – that urticates within minutes of contact with the offending allergen (Fig. 17.1).

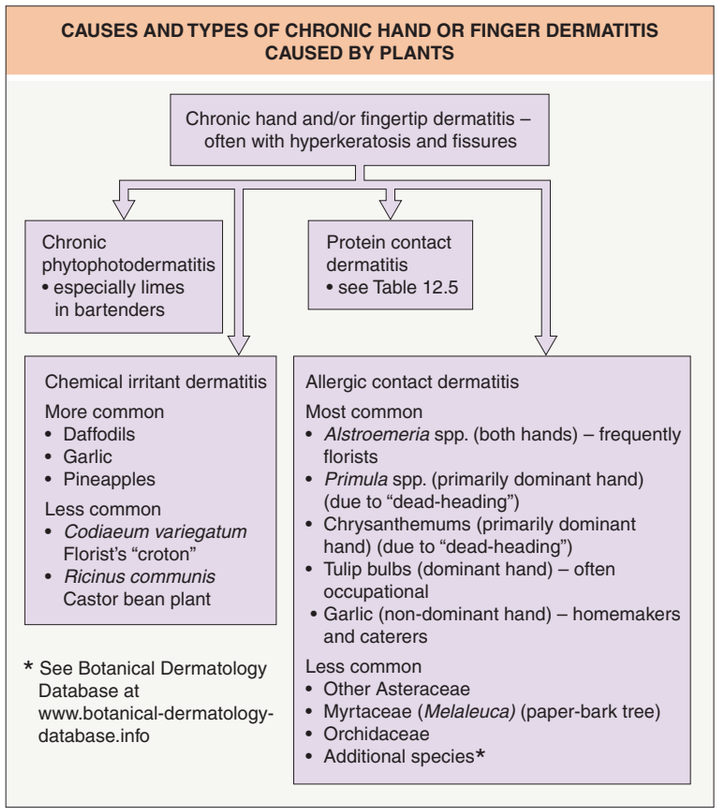

Fig. 17.1 Causes and types of chronic hand or finger dermatitis caused by plants.

Differential Diagnosis and Pathology

See Toxin-Mediated (Non-immunologic) Contact Urticaria and Chs. 16 & 18.

Treatment

Prevention is the preferred form of "treatment". Parenterally administered epinephrine (adrenaline) is required for anaphylactic reactions.

TOXIN-MEDIATED (NON-IMMUNOLOGIC) CONTACT URTICARIA

Key features

- Anyone can be affected
- Stinging nettles (*Urtica* spp.) are most common cause
- Reaction can be subjective only
- Sharp hairs on the plants can contain histamine, serotonin, and acetylcholine

History

Plants causing toxin-mediated urticaria have been used since antiquity as counterirritants in folk medicine, and Native Americans used stinging nettles to treat rheumatism, stomach upset, postpartum hemorrhage, paralysis, fevers, colds, and tuberculosis. Stinging nettle is used to produce homemade diuretics, and stem fibers were even used to make cloth until the early twentieth century.

Epidemiology

Urticaceae family members cause the majority of plant-induced contact urticaria. Because everyone exposed to the toxins develops urticaria, toxin-mediated urticaria is far more common than immunologic urticaria. Since few affected people seek medical attention, the true incidence of toxin-mediated urticaria is unknown. The most common culprit in the US is the stinging nettle (*Urtica dioica*; Fig. 17.2A)[7], which is widely scattered throughout the northern hemisphere (except in lowland tropical areas), especially in moist woods, roadsides, and on wasteland. Other common urticating plants are listed in Table 17.4.

Pathogenesis (see Ch. 18)

Inciting plants possess sharp hairs (trichomes) on leaves and stems (Fig. 17.2B,C). The proximal silicaceous hair is attached to a distal calcified portion that possesses a terminal bulb. When rubbed against, the bulb dislodges to reveal a beveled, hypodermic needle-like, hollow hair (Fig. 17.2D). The latter releases an irritant chemical cocktail (histamine, acetylcholine, serotonin) that supposedly serves as a defense mechanism against herbivores.

Clinical Features

Wheals achieve maximal size 3 to 5 minutes after contact, and erythema, burning, and pruritus last 1–2 hours. Paresthesias may last 12 hours or more. Although histamine, acetylcholine, and serotonin explain the early cutaneous reaction, they do not account for the persistent paresthesias.

Reactions to stinging nettle pale in comparison to those elicited by members of the *Dendrocnide* genus of the Urticaceae family. These trees grow up to 40 meters tall in eastern Australian rainforests. Young

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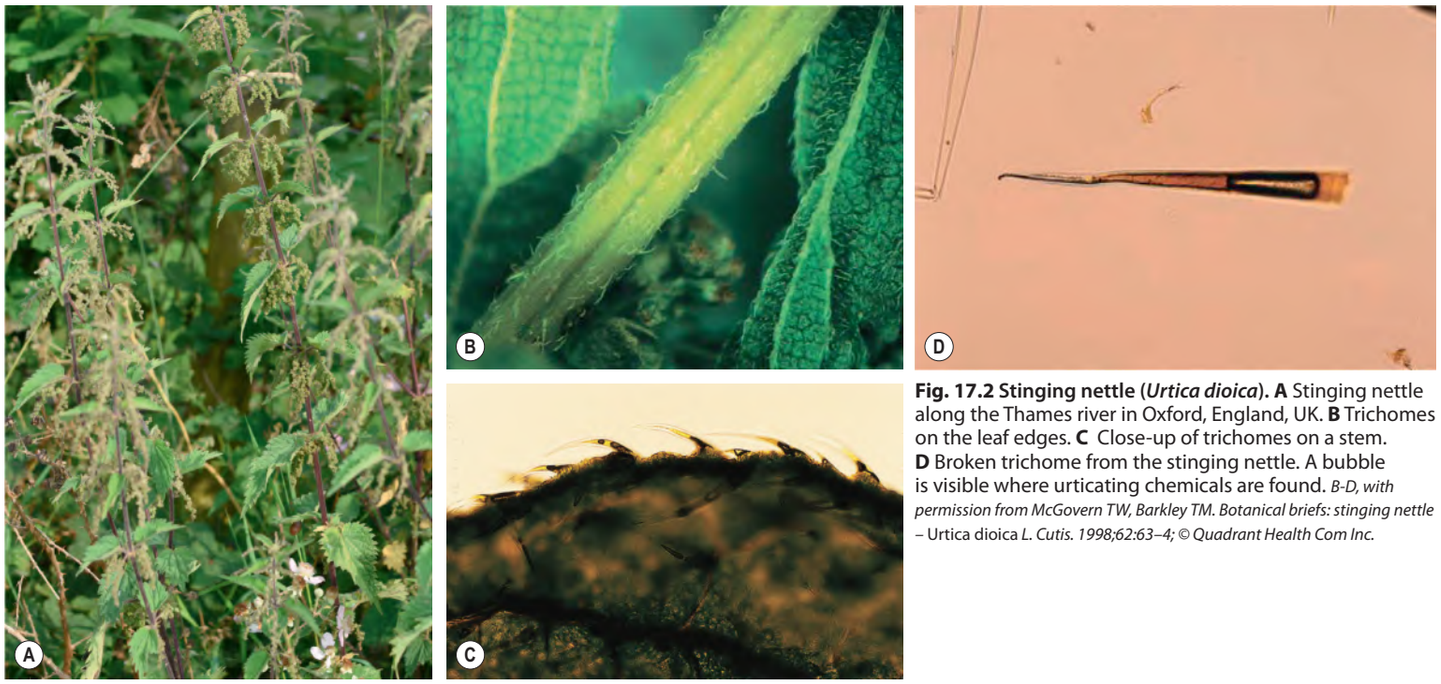

Fig. 17.2 Stinging nettle (*Urtica dioica*). A Stinging nettle along the Thames river in Oxford, England, UK. **B** Trichomes on the leaf edges. **C** Close-up of trichomes on a stem. **D** Broken trichome from the stinging nettle. A bubble is visible where urticating chemicals are found. *B-D, with permission from McGovern TW, Barkley TM. Botanical briefs: stinging nettle – Urtica dioica L. Cutis. 1998;62:63–4; © Quadrant Health Com Inc.*

THE MOST COMMON PLANTS CAUSING TOXIN-MEDIATED URTICARIA			
Family	**Genus**	**Species**	**Notes**
Urticaceae (nettle family)	*Urtica*	*dioica, urens, pilulifera*	Stinging nettles – worldwide
	Laportea	*canadensis*	"Wood nettle", 5-foot-tall perennial herb – NE USA
	Dendrocnide	*gigas, moroides, photinophylla*	Potentially deadly stinging trees – eastern Australian rainforests
Euphorbiaceae (spurge family)	*Acidoton*	*urens*	Tropical Americas
	Cnidoscolus	*stimulosus*	Spurge nettle – SE USA
		other species	Spurge nettles of tropical Americas
Hydrophyllaceae (water-leaf family)	*Wigandia*	*caracasana, urens*	Large-leafed shrubs – tropical Americas

Table 17.4 The most common plants causing toxin-mediated urticaria.

shoots are covered with stiff stinging hairs. Severe urticaria from these may last for weeks, and contact with water or cold objects reactivates the urticaria. Severe, intermittent, stabbing pains may follow the course of lymphatics. Human and animal deaths due to *Dendrocnide* contact have been documented.

Pathology (see Ch. 18)

Five minutes after contact with *Urtica dioica*, dermal edema and telangiectasias with or without mild spongiosis are seen. At 12 hours, the edema resolves but vasodilation persists. Some patients develop spongiosis in conjunction with a neutrophilic and/or lymphocytic infiltrate. Mast cells are observed within the papillary dermis at 12 hours, but not at 5 minutes.

Differential Diagnosis (see Ch. 18)

Although the evaluation of such patients is difficult, office testing of suspected allergens or toxin-containing plants may involve one of several methods[4]. The most sensitive tests for immunologic contact urticaria are the prick and scratch-chamber tests. For the scratch-chamber test, a 5-mm scratch is made on the back or forearm. Test material is applied and occluded with a Finn chamber for 15 minutes. The site is examined following Finn chamber removal and every 15 minutes for an hour. After reading, the chamber can be replaced for 48 hours to test for delayed hypersensitivity.

The open application test provides the most reliable way to test for toxin-mediated urticaria. Samples (0.1 ml) from a series of dilutions are each spread onto discrete 3 × 3 cm areas of skin. Sites are observed every 10–15 minutes for an hour. Maximal erythema and edema typically occur 30–40 minutes after application.

Treatment

Most stings are benign, self-limited, and require no treatment. Trichomes may be removed with glue and gauze as described in the next section. Topical pramoxine or oral analgesics may provide some symptomatic relief.

MECHANICAL IRRITANT DERMATITIS

Key features

- Large spines (modified leaves) and thorns (modified branches) cause penetrating injuries and secondary infection
- Small fishhook-shaped spines (glochids) can embed in the skin
- Prickly pears are a cause of glochid dermatitis

Epidemiology

Mechanical irritant dermatitis can affect anyone. Many plants, including most cacti, can inflict mechanical injury via small or large emergences. Generally, the amount of damage to the skin is inversely proportional to the size of the emergence. Many of the plant families that can cause mechanical irritant dermatitis are listed in Table 17.5.

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COMMON PLANTS KNOWN TO CAUSE MECHANICAL IRRITANT REACTIONS		
Family	**Genus species**	**Notes**
Amaranthaceae	*Kali tragus*	Tumbleweed or Russian thistle – sharp-edged spines penetrate skin
Araliaceae	*Hedera helix*	Common ivy – leaf-borne stellate hairs detach as leaves age
Asteraceae	*Carduus* and *Cirsium* spp.	Thistles
	Lactuca serriola	Prickly lettuce
Boraginaceae	*Borago officinalis*	Borage – sharp, stiff leaf and stem hairs
Cactaceae	*Opuntia* spp.	Prickly pears; "Sabra dermatitis"
Liliaceae	*Tulipa* spp.	Coarse fibers on tulip bulb tunics cause irritant component of "tulip fingers"
Moraceae	*Ficus* spp.	Figs – abrasive bristles on leaves and fruit
	Morus spp.	Mulberries – abrasive bristles on leaves and fruit
Nyctaginaceae	*Bougainvillea*	Sharp spines on stems and spear-shaped crystals in trichomes on leaves (Brazil native)
Poaceae	Many spp.	Grasses – fine hairs, prickly spikes, and cutting leaf edges
Proteaceae	*Grevillea* spp.	Terminal, sharp points on leaves – Australia
Rosaceae	*Rosa* spp.	Thorns can cause penetrating injuries, tenosynovitis, and foreign body granulomata
Rubiaceae	*Galium aparine*	"Goosegrass" – hooked prickles on fruit, stems, leaves
Scrophulariaceae	*Verbascum thapsus*	"Mullein" or "flannel-plant" – woolly hairs on leaves used as facial rubefacient
Sterculiaceae	*Fremontodendron* spp.	"California glory" possesses stiff, stellate hairs

Table 17.5 Common plants known to cause mechanical irritant reactions. Although numerous plants possess irritant appendages, some of the well-known ones are listed here.

Pathogenesis

Whereas cacti possess large spines, their smaller glochids cause more notorious dermatologic problems. The glochids – tufts of hundreds of short, barbed or hooked hairs – arise from pincushion-like structures called "areoles", from which larger spines may also arise. The minute, barbed glochids (Fig. 17.3A) often point outward and backward like a fishhook and produce considerable irritation and pruritus after penetrating the skin. For example, *Opuntia microdasys* ("polka dot cactus", "bunny ears cactus"), a house and garden favorite, bears disarming-appearing "fluffy" clusters of 100–200 glochids on its pads (see Fig. 17.12).

Clinical Features

Prickly pear glochids from *Opuntia* spp. cause "Sabra dermatitis", a pruritic, papular eruption occurring among prickly-pear pickers and those who unwarily stumble into burglar-proof hedges of this native Mexican plant (Fig. 17.3B). The fruit contains the highest concentration of glochids (Fig. 17.3C), and the ensuing eruption can resemble fiberglass dermatitis or scabies. Prickly pears should be picked only when wetted, and harvesting should cease when it is windy, since the glochids can become airborne.

All 200 or so species of *Opuntia* are native to the New World, ranging from New England and British Columbia southward to the Straits of Magellan. Numerous species have become established in the Mediterranean basin, South Africa, South Asia, and Australia.

Spine and thorn injuries can be complicated by cutaneous inoculation of microorganisms such as *Clostridium tetani* and *Staphylococcus aureus*. Grasses, sphagnum moss, and rose thorns can transmit *Sporothrix schenckii*. Atypical mycobacteria such as *Mycobacterium kansasii* (blackberries), *M. marinum* (cactus spines), and *M. ulcerans* (spiky tropical vegetation) have also added infectious insult to mechanical injury.

Treatment

In a controlled study in rabbits, *Opuntia* glochids were most effectively removed by first detaching the larger clumps with tweezers and then applying glue and gauze to the affected area. After the glue dried, the gauze was grasped and peeled off, resulting in the removal of 95% of implanted glochids[8].

CHEMICAL IRRITANT DERMATITIS

Key features

- Calcium oxalate in daffodils is a major cause of plant-related irritant dermatitis in florists and horticulturalists
- Spurges (including poinsettias) contain irritant phorbol esters in latex
- Capsaicin in hot peppers affects cutaneous nerves, but not the skin

Fig. 17.3 Prickly pears (*Opuntia* spp.) and their glochids. A Microscopic appearance of glochids from *Opuntia microdasys*. **B** Prickly pear (*Opuntia ficus-indica*) habit. **C** Prickly pear (*Opuntia ficus-indica*) fruit. Fine spines and glochids are visible. *Courtesy Dirk Elston, MD.*

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History

Plant-derived irritant chemicals have been recognized for centuries. Native Mexican Indians once used smoke from burning peppers (*Capsicum* [Solanaceae]) as a weapon against enemies, and latex from the manchineel tree (*Hippomane mancinella* [Euphorbiaceae]) was used by Caribbean people as poison for arrow tips.

Epidemiology

Several major plant families found throughout the world contain chemical irritants (Table 17.6).

Pathogenesis, Clinical Features, and Treatment (see Ch. 15)

Calcium oxalate, one of the most common chemical irritants, is found in *Dieffenbachia picta* (Araceae), a plant that decorates millions of homes and public places. Upon contact with a moist surface, the leaves release water-insoluble calcium oxalate. When leaves are chewed, salivation, burning, mucosal edema, and blistering follow. This causes hoarseness or aphonia, hence the common name "dumb cane". Treatment includes parenteral corticosteroids, antacid mouthwashes, and analgesics; antihistamines are of no benefit. The pain and edema wane over 4–12 days.

Calcium oxalate enhances the irritancy of other chemicals, including the proteolytic enzyme bromelain (bromelin) found in pineapples. Pineapple workers often develop cracks, fissures, fingerprint loss, and microhemorrhages on their hands. Calcium oxalate-induced microabrasions allow bromelain to exert its proteolytic effect on dermal blood vessels. Nitrile gloves can be worn for protection.

Bulb dermatitis, though less dramatic, is more common[9]. Daffodil sap, found not only in bulbs but also in stems and leaves, contains calcium oxalate and causes "daffodil itch", probably the most common dermatitis in florists. Dryness, scaling, fissures, and erythema develop on the fingertips, hands, and forearms. Bulbs of several other plants (e.g. tulips, hyacinths) also commonly cause dermatitis. Allergens found in garlic bulbs act as irritants at high concentrations leading to second- and third-degree burns as well as subungual hyperkeratosis and hemorrhage that can mimic nail psoriasis.

While spurges (Euphorbiaceae) are notorious for their irritant latex, they derive their common name from the purgative properties of their seeds. The milky latex contains irritant phorbol esters, diterpenes, and daphnane esters that may cause a painful, blistering dermatitis and even temporary blindness if the latex contacts the eye. Sitting under a manchineel tree in the Caribbean during a rainstorm will lead to intense cutaneous burning, erythema, and bullae, often accompanied by periocular swelling through transfer of the latex by rain. Manchineel fruit is known locally as the "apple of death". The latex of the best-known family member, the poinsettia (*Euphorbia pulcherrima*), is only mildly irritating.

Buttercups (Ranunculaceae) contain the glycoside ranunculin that is converted to protoanemonin after plant injury. Protoanemonin causes severe, linear vesiculation that may resemble early phytophotodermatitis, but no hyperpigmentation ensues. Because protoanemonin rapidly polymerizes to the non-irritant anemonin, only freshly damaged plants cause reactions.

"Chili burns" commonly occur among those who remove skins from large batches of roasted chili peppers (*Capsicum annuum*). When applied to skin, the active principle, capsaicin, depolarizes nerves causing vasodilation, smooth muscle stimulation, glandular secretion, and sensory nerve activation. Because only nerves are affected, there is erythema but no vesicles or bullae. Symptoms may be delayed and last hours to days. The best home remedy for chili burns is handwashing with soap and water followed immediately by 1 hour of immersion in vegetable oil to remove the fat-soluble capsaicin.

The "century plant" (*Agave americana*) grows in temperate, subtropical, and tropical climates and forms rosettes of tough, sword-shaped leaves with spiny margins. The marked pruritus and stinging that follows within five minutes of contact with the agave leaf is thought to be caused by latex-derived calcium oxalate crystals and saponins. Purpura may appear subsequently. Soldiers have been known to try to "earn" sick leave by rubbing broken leaves on their skin to induce a severe vesiculopustular irritant dermatitis, often accompanied by systemic symptoms[10].

PLANTS MOST COMMONLY IMPLICATED IN CAUSING CHEMICAL IRRITANT DERMATITIS			
Family	**Binomial name**	**Common name**	**Irritant chemical (location)**
Agavaceae	*Agave americana*	Century plant	Calcium oxalate, saponins (latex throughout plant)
Alliaceae	*Allium sativum*	Garlic	Thiocyanates (bulbs)
Amaryllidaceae	*Narcissus* spp.	Daffodil, narcissus	Calcium oxalate (stems, leaves, bulbs)
Anacardiaceae	*Anacardium occidentale*	Cashew tree	Cashew nut shell oil (mesocarp [middle layer] of nut shell [irritant and allergenic])
Araceae	*Dieffenbachia picta*	Dumb cane	Calcium oxalate (leaves and fruits)
	Philodendron spp.	Philodendron	Calcium oxalate (leaves)
Brassicaceae	*Brassica nigra*	Black mustard	Thiocyanates (all parts)
	Raphanus sativus	Radish	
Bromeliaceae	*Ananas comosus*	Pineapple	Bromelain (stem > fruit) Calcium oxalate (all parts)
Euphorbiaceae	*Codiaeum variegatum*	Florist's "croton"	Phorbol esters (latex in stems, leaves, fruits)
	Euphorbia esula	Leafy spurge	
	Hippomane mancinella	Manchineel tree	Standing in rain beneath manchineel tree can transfer highly irritant latex and even cause blindness
	Euphorbia pulcherrima	Poinsettia	
Liliaceae	*Hyacinthus orientalis*	Hyacinth	Calcium oxalate (bulbs)
Polygonaceae	*Rheum rhaponticum*	Rhubarb (false, rhapontic)	Calcium oxalate (leaves)
Ranunculaceae	*Ranunculus* spp.	Buttercups	Protoanemonin (freshly damaged plant parts)
	Aquilegia spp.	Columbine	
	Caltha spp.	Marsh marigold	
Solanaceae	*Capsicum annuum*	Hot pepper	Capsaicin (placenta of fruit [no dermatitis, only erythema, edema, burning])

Table 17.6 Plants most commonly implicated in causing chemical irritant dermatitis.

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PHYTOPHOTODERMATITIS

Key features

- Non-immunologic – it can happen to anyone
- Need UVA light plus topical or oral contact with photosensitizer
- Furocoumarins (psoralens and angelicins) are the most common photosensitizers
- Limes, celery, and rue (herb-of-grace) are the most common causes
- Potential photosensitizing plants of the family Apiaceae can be identified by umbel flower structure

Phytophotodermatitis connotes phototoxic reactions consisting of erythema (with or without blistering) and delayed hyperpigmentation following exposure to plants. Phototoxicity implies an inflammatory reaction caused by the combination of a topical or oral photosensitizing agent followed by exposure to the appropriate wavelength of UV radiation. Because this is not an immunologic reaction, no prior sensitization is necessary and anybody can be affected.

History

As early as 2000 BCE, vitiligo patients were encouraged to lie in the sun after picking *Ammi majus* from the Nile River valley and rubbing the juice on their skin. By 1400 BCE, Indian doctors were using boiled extracts of *Psoralea corylifolia* (scurf-pea, babchi), from which we derive the term "psoralen". Even nowadays in India, its seeds remain a source of psoralens for the treatment of vitiligo.

In 1897, Apiaceae-induced dermatitis was first reported, but the need for concomitant UV radiation went unnoticed. It would be another 40 years before Klaber introduced the term "phytophotodermatitis" to emphasize that a combination of plants ("*phyto-*") plus light ("*photo-*") is required for the reaction.

Epidemiology

Apiaceae (formerly Umbelliferae) and Rutaceae (citrus family) plants are the most common causes of phytophotodermatitis (Table 17.7 & 17.8). Members of the mulberry (Moraceae) and pea (Fabaceae) families also contain furocoumarins.

Apiaceae

A distinctive floral umbel makes family members easy to recognize: numerous small flowers form a simple umbel (a cluster of flowers on stalks of roughly equal length arising from a single point). Many small umbels can form "compound umbels" (Fig. 17.4A). Compound umbels may be umbrella-shaped or ball-shaped. The flower heads are sheathed at the base by one or more leaf-like bracts.

Common hogweed, also referred to as cow parsnip, is known by three species names: *Heracleum lanatum*, *H. maximum*, or *H. spondylium*. In North America, it naturally grows over 2 meters tall (Fig. 17.4B,C), with leaves up to 1 meter long. The larger giant hogweed (*H. mantegazzianum*) is a European native that has now invaded parts of North America. Both cause phytophotodermatitis. Fruits contain the highest concentrations of psoralens followed by leaves and stems. The greatest threat from hogweed occurs in the autumn, when the weather favors the development of many seeds. One severe case of occupational contact with giant hogweed during mid-summer roadside lawn mowing led to marked phytophotodermatitis-induced soft tissue necrosis of the leg and eventual below-the-knee amputation.

Rutaceae

The Rutaceae family includes tropical (*Citrus* spp.), subtropical, and temperate (*Ruta* spp.) plants and is the second most common plant family (after Apiaceae) that causes phytophotodermatitis (see Table 17.8). Many Rutaceae grow as shrubs or small trees and produce fleshy fruit. The rind of the Persian lime (*Citrus latifolia*), which contains ten times more 5-methoxypsoralen (5-MOP) than the pulp, is a major cause of phototoxic reactions in the US, especially in Florida and the desert Southwest. Outdoor bartenders commonly develop vesicles on their index and middle fingers from squeezing limes to make cocktails. Even the sweet orange (*Citrus sinensis*) can cause a phototoxic cheilitis after contact with the rind.

Garden rue (*Ruta graveolens*) is a bitter tasting and smelling sub-shrub native to the Mediterranean basin with a long history as a folk medicine

IMPORTANT MEMBERS OF THE APIACEAE FAMILY (FORMERLY UMBELLIFERAE) IMPLICATED IN PHOTOTOXIC REACTIONS		
Scientific name	**Common name(s)**	**Comments**
Ammi majus	False Bishop's weed	Historical and economic importance
Angelica archangelica	Angelica	Candied portions used in cakes; characteristic Benedictine flavor
Angelica sylvestris	Wild angelica	
Anthriscus sylvestris	Cow parsley, wild chervil	"Strimmer dermatitis"*
Apium graveolens	Celery	Fungus-infected plants produce more psoralens
Ficus carica	Fig	Widespread second-degree burns when used as tanning promoter (Turkey)
Foeniculum vulgare	Fennel	
Heracleum laciniatum	Tromsø palm	Scandinavia
Heracleum lanatum, H. maximum, H. sphondylium	Cow parsnip (common hogweed)	North America; most common cause of "strimmer dermatitis"*
Heracleum mantegazzianum	Giant (Russian) hogweed, wild rhubarb, hogweed tree	Giant weed naturalized and invasive across UK, Europe, Canada, US
Heracleum persicum	Persian hogweed	Iran, Iraq, Turkey
Notobubon galbanum	Blister bush	South Africa
Pastinaca sativa	Parsnip	Furocoumarins in leaves, roots, stem, fruits
Petroselinum crispum	Parsley	
*Strimmer = string trimmer, a garden tool for cutting weeds.		

Table 17.7 Important members of the Apiaceae family (formerly Umbelliferae) implicated in phototoxic reactions.

PHOTOTOXIC MEMBERS OF THE FAMILY RUTACEAE	
Scientific name	**Common name(s)**
Citrus aurantiifolia	Key lime
Citrus latifolia	Persian lime or seedless lime
Citrus aurantium	Bitter orange
Citrus bergamia	Bergamot orange
Citrus sinensis	Sweet orange
Citrus limetta	Sweet lemon
Citrus limon	Lemon
Citrus paradisii	Grapefruit
Cneoridium dumosum	Coast spice bush, berry rue
Dictamnus albus	Burning bush, gas plant
Pelea anisata	Mokihana (berries used for lei in Hawaii)
Phebalium squamulosum	Blister bush, common in Western Australia
Ruta graveolens	Rue, herb-of-grace

Table 17.8 Phototoxic members of the family Rutaceae.

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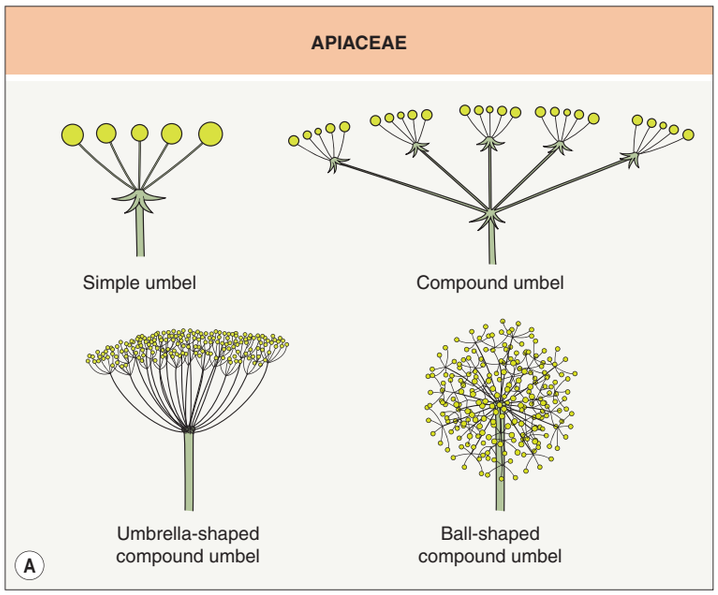

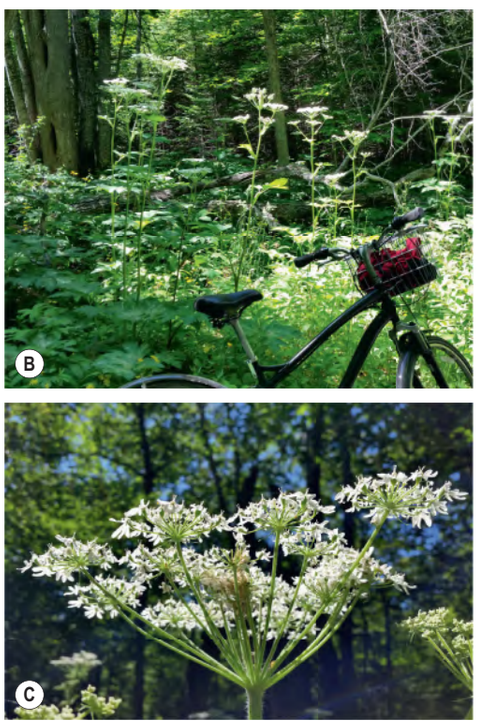

Fig. 17.4 Apiaceae. **A** Compound umbels can be umbrella-shaped or ball-shaped. **B** Cow parsnip (common hogweed; *Heracleum maximum*) habit, Mackinac Island, MI, USA. **C** Compound umbel of cow parsnip (common hogweed)

Fig. 17.5 Bullous phase of phytophotodermatitis. There was associated burning but no pruritus, and the linear bullae were replaced by hyperpigmentation. *Courtesy Jean L. Bolognia, MD.*

remedy and insect repellant. It contains 5-MOP, 8-MOP, and angelicins and is probably the most common cause of phototoxicity acquired in English gardens.

Dictamnus albus, the gas plant or burning bush, exudes an aromatic oil that can be ignited briefly without harming the plant. It has become a common yard plant in the US and Canada, and it grows wild in central and southern Europe, eastern Siberia and northern China. The seed pods have high concentrations of both 5-MOP and 8-MOP.

Moraceae

The mulberry family includes the fig tree (*Ficus carica*), a Middle Eastern native that has been widely cultivated throughout warm, temperate regions of the world. Psoralens are found chiefly in the sap of leaves and shoots but not in the fruit[11].

Other families

Eleven of 23 HIV-infected patients receiving systemic hypericin (from St John's wort, *Hypericum perforatum* [Hypericaceae]) as a potential antiretroviral agent developed severe cutaneous phototoxicity.

Pathogenesis

Furocoumarins possess linear (psoralens) or angular (angelicins) tricyclic structures (see Fig. 134.9). Bergapten (5-MOP), originally isolated from the southern European bergamot orange (*Citrus bergamia*), and 8-MOP (xanthotoxin), first isolated from the Nigerian prickly ash (*Fagara zanthoxyloides*), cause the most severe reactions.

In the presence of UVA (peak effectiveness at 320–340 nm), psoralens within the skin can interact with molecular oxygen and form reactive oxygen species that induce cytoplasmic vacuolization and membrane rupture of keratinocytes. Within 2 hours of UVA exposure, desmosomal plaques detach and degenerate leading to blister formation without the need for DNA damage[12].

UVA also excites intercalated psoralens to form covalent interstrand DNA cross-links by binding to pyrimidines (see Fig. 134.10). This leads to further keratinocyte cell death (apoptosis, "sunburn cells") and stimulates hyperpigmentation via increased melanocyte mitosis and dendricity, melanocyte hypertrophy, increased tyrosinase activity, larger melanosomal size, and enhanced distribution of melanosomes throughout the epidermis.

Furocoumarins appear necessary to defend plants from fungal attack. Healthy celery plants (*Apium graveolens*) contain 10–100 mcg/g wet weight of psoralens, but they may produce 320 mcg/g when infected with pink-rot fungus (*Sclerotinia sclerotiorum*). Disease-resistant celery unfortunately possesses high levels of furocoumarins concentrated in the leaf and stem sap.

Clinical Features

Cutaneous sensitivity to UVA light peaks 30–120 minutes after contact with furocoumarins. Bizarre configurations of erythema, edema, and bullae appear after 24 hours and peak at 72 hours (Fig. 17.5). These painful, non-pruritic reactions are more often seen in mid to late summer, when psoralen concentrations are highest in the offending plants and more skin is exposed to direct sunlight. To avoid misdiagnosis as poison ivy dermatitis, physicians must note that the initial erythematous and bullous reaction only occurs in sun-exposed areas. Hyperpigmentation appears 1–2 weeks later and lasts months to years (Fig. 17.6). Occasionally, low-dose UVA and/or psoralens cause hyperpigmentation without a preceding vesicular or erythematous eruption (see Ch. 67). Involved areas may remain hypersensitive to UV radiation for years. Wet skin, sweating, and heat enhance the phototoxic response.

Furocoumarin exposure occurs in many settings (Table 17.9). Modern power tools such as weed-whackers and strimmers (string trimmers) deliver a buckshot spray of weeds that may include phototoxic plants. "Strimmer dermatitis" appears 12–24 hours later as red, irregular

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macules and papules, not bizarre linear and angular streaks, on the chest, arms, and legs if wearing shorts. Implicated species, especially in Europe, include common hogweed (cow parsnip; *Heracleum sphondylium*), giant hogweed (*Heracleum mantegazzianum*), and cow parsley (*Anthriscus sylvestris*). Berlock (or berloque, meaning trinket or charm) dermatitis is characterized by pendant-like streaks of pigmentation on the neck, face, arms, or trunk after the application of colognes containing 5-MOP.

Fig. 17.6 Phytophotodermatitis. Streaky hyperpigmentation due to contact with a psoralen-containing plant followed by exposure to sunlight. *Courtesy Lorenzo Cerroni, MD.*

Phytophotodermatitis has also been mistaken for child abuse (see Ch. 90). Parents who touch plants or fruit containing furocoumarins can transfer these chemicals to their children. A week or more later, the children may develop digitate hyperpigmentation in sun-exposed sites of parental hand contact. Multiple hues are not seen in the skin lesions, as in healing bruises. In addition, lime-induced phytophotodermatitis can be misdiagnosed as allergic contact dermatitis, impetigo, cellulitis, and infectious lymphangitis.

Celery harvesters and canners are at high risk for developing phytophotodermatitis. Only 1 mcg of 8-MOP/cm^2 of skin is necessary to produce blisters after 2.4 J/cm^2 (less than 10 minutes) of summer sunlight in Colorado. In one study of 320 randomly selected Michigan celery workers, 163 (51%) displayed various stages of vesicular and bullous dermatitis on the fingers, hands, and forearms[13]. In this study, the authors could not induce phytophotodermatitis except by using celery infected with pink-rot fungus. However, phototoxicity following the ingestion of naturally occurring psoralens is uncommon.

Phytophotoallergic Contact Dermatitis

Photoallergy due to plants is rare. A 20-year review of photopatch testing results from a tertiary medical center found reactions in eight of 69 patients, with reactions in only four patients thought to be clinically relevant[14].

Treatment

Prevention is the best treatment for phytophotodermatitis. Known furocoumarin-containing plants should not be planted near play areas. When using weed-whackers or strimmers, operators must cover their chest and extremities. If one has been in contact with a suspected phototoxic plant, prompt washing with soap and water may prevent a reaction.

ALLERGIC CONTACT DERMATITIS (SEE CH. 14)

Members of the Anacardiaceae and Asteraceae families represent the most common causes of allergic contact dermatitis due to plants.

RISK ACTIVITIES FOR PHYTOPHOTODERMATITIS	
Cosmetics	Tan promoters or perfumes containing bergamot oil (berlock [berloque] dermatitis) or a fig leaf decoction
Fruit and vegetable processing	Canning celery or stocking celery in grocery stores Making lemonade or limeade, especially if selling it outside Squeezing lime juice for margaritas and other drinks or guacamole Applying wild carrot decoction to treat foot swelling (China)
Gardening	Brushing against *Dictamnus* spp. ("gas plant/burning bush") (US, Europe, N China) or *Ruta* (UK) Cultivating celery, parsnip, or parsley Clearing weeds with a "weed whacker" (US) or "strimmer" (UK) Pruning or harvesting figs Growing *Angelica* for herbal medicine (Korea), cake decorating (when candied), tonic and flavoring in wines (esp. Benedictine, US)
Hiking	Through fields and riverbanks (*Heracleum* spp.) (Pacific NW, Europe) Hiking in southern California and Baja California (*Cneoridium dumosum*, coast spice bush [Rutaceae])
Ingestion	Ingestion of massive quantities of psoralens (esp. celery) before UVA tanning Eating selemez (wild spinach, *Chenopodium album* [Chenopodiaceae]) – native in Europe and Asia, cultivated widely; sensitizing chemical unknown
Medications	Excessive exposure to UV radiation after taking or applying psoralens for PUVA Application of rue (*Ruta* spp.) as an insect repellant
Play	Making peashooters with *Heracleum* spp. Playing among rue bushes or Apiaceae (see Table 17.7) Using psoralen-containing plants as "weapons" Wearing lei of *Pelea anisata* (Hawaii)

Table 17.9 Risk activities for phytophotodermatitis.

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Epidemiology

Anacardiaceae and related families

Key features

- Poison ivy/poison oak plant identification
  - Compound leaves with three leaflets
  - Flowers or fruit arise from axillary position
  - Black dots of urushiol often present on leaves and fruit
  - Common poison ivy climbs as a vine with hairy aerial roots
- Allergens are long-chain catechols and resorcinols found in the urushiol oleoresin

Members of the Anacardiaceae family cause more allergic contact dermatitis than all other plant families combined. Most allergenic family members belong to the genus *Toxicodendron* (meaning "poison tree"). In the US, over a third of patients present during June or July[15].

Toxicodendron identification

Compound *Toxicodendron* leaves possess three or more odd-numbered leaflets[16]. Flowers and fruit arise in an axillary position in the angle between the leaf and the twig from which it arises (Fig. 17.7A,B). The leaf stalk is enlarged at its origin from the supporting twig and leaves a "U"- or "V"-shaped scar after it falls off. Green fruit turn off-white when mature. Plants cling to trees via "hairy" aerial rootlets (Fig. 17.7C,D).

Toxicodendron oleoresin contains urushiol and the enzyme laccase which oxidizes urushiol to form an insoluble, black, plastic-like polymer with great tensile strength. Black spots are commonly found on the leaves of plants (Fig. 17.7E), and the "black-spot test" helps identify toxic Anacardiaceae, but performing it also increases one's likelihood of contracting a nasty dermatitis! A stone is used to thoroughly crush plant contents, especially the leaf stalks, between folds of white paper. Urushiol turns dark brown within 10 minutes and black by 24 hours.

Allergenic Anacardiaceae

As noted in Table 17.10, there are two species each of poison ivy and poison oak and one species of poison sumac that are common to the US (Fig. 17.8). Some have multiple subspecies. Poison oak and poison ivy are weeds that grow along roads, trails, or streams; they possess three (or occasionally five) leaflets per leaf. Poison sumac contains 7–13 leaflets per leaf (see Fig. 17.7A). Young leaves are frequently red in color, and the mature fruit (drupes) are tan or cream-colored and hairless (young fruit has hair). Classically, poison ivy leaves have pointed tips and are ovate (widest point below the center). Poison oak leaves usually have rounded ends. Western poison oak has oval leaves, whereas Eastern poison oak has variable leaf appearances that can mimic white oak leaves.

Virtually all portions of the plant can induce dermatitis, even in the winter. The "winter" shape of poison ivy on fence posts resembles "Medusa heads" that are identifiable while driving down the highway. Strongly suspect any vine climbing by hairy aerial rootlets that is connected at the bottom to a tree or log. In snow, Western poison ivy grows so low to the ground that the upward, tine-like branches may be the only clue to its identification.

The cashew nut tree (*Anacardium occidentale*) grows in the tropics worldwide. The nut contains an oily, brown juice between the two layers of its shell. Contact with concentrated phenols in the nutshell and bark leads to immediate vesication. Any part of the tree, except the nut, can cause dermatitis.

Mango (*Mangifera indica*) is the most popular fruit tree in tropical and subtropical America, and some 35 species grow throughout Southeast Asia. The leaves, bark, stems, and fruit skin contain sensitizing resorcinols. Peeling the fruit before eating it typically prevents allergic contact dermatitis, although allergy to mango pulp has been demonstrated; the pulp allergens appear to be distinct from the resorcinols. Hawaiian natives rarely react to mango, perhaps because early oral exposure induces immunologic tolerance.

The Florida holly (*Schinus terebinthifolius*, Brazilian pepper tree) is probably the most common cause of allergic contact dermatitis in southern Florida. The sap and crushed berries possess a variety of sensitizing catechols and resorcinols.

Although the mango, cashew, and Brazilian pepper trees are incredibly common throughout Latin America, dermatitis to them there is rare, in contrast to the frequent reactions seen in US residents. The main allergens in toxicodendrons are catechols, whereas those in the listed trees are resorcinols, and catechols possess greater allergenicity than resorcinols. Early *cutaneous* exposure to catechols may allow

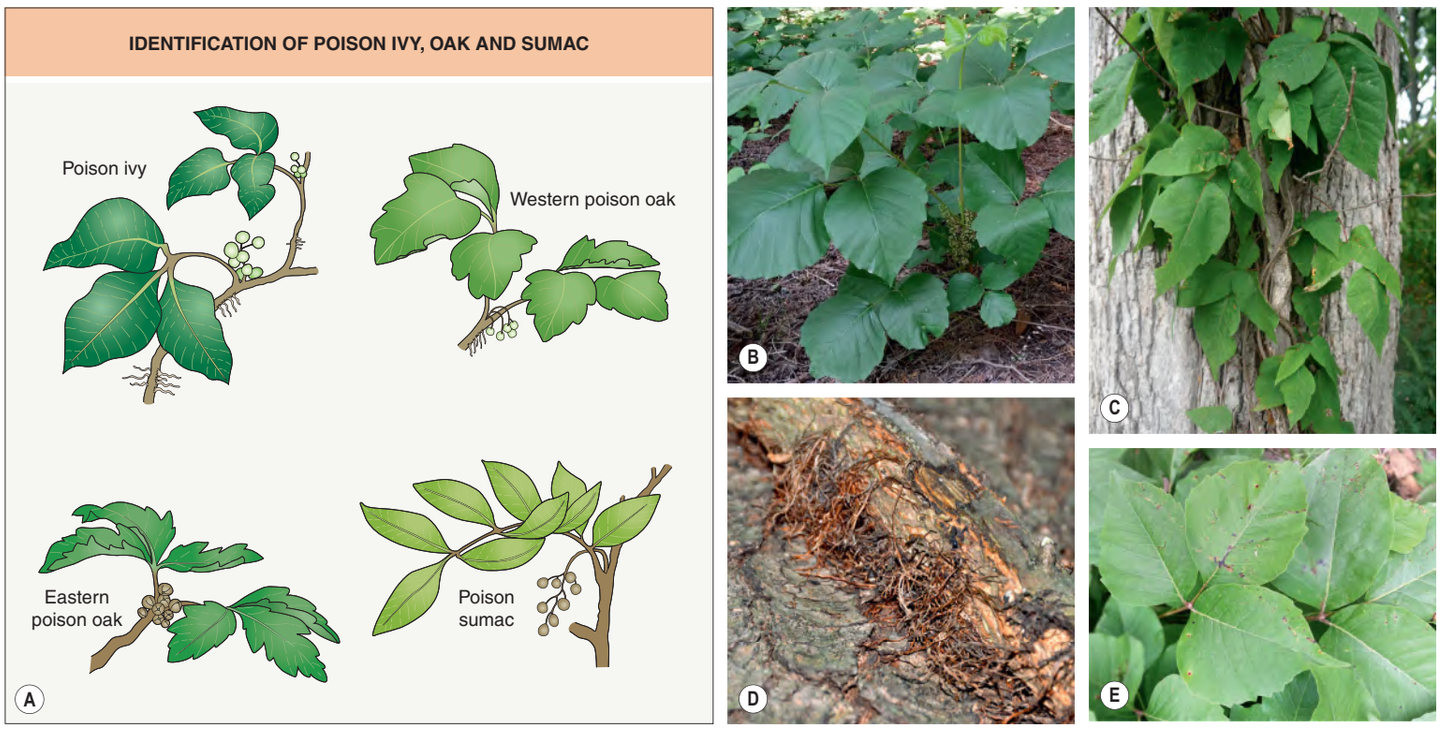

Fig. 17.7 Characteristic features useful for identifying poison ivy, poison oak and poison sumac. A Shapes and numbers of leaflets and appearance of fruit (drupes). **B** Shrub type poison ivy (*T. radicans*) with flowers arising from an axillary position just below point where leaf attaches to stem. Leaves possess three leaflets ("*Leaves of three, let them be*" or "*One, two, three, don't touch me*"). **C, D** Climbing vine of poison ivy (*T. radicans*) on an oak tree demonstrating aerial roots that anchor the vine to the tree; close-up of vine anchored to tree bark by fine hair-like aerial roots ("*Hairy rope, don't be a dope*"). **E** Shrub type poison ivy (*T. radicans*) with black spots on leaflets. Leaf trauma (e.g. wind, rain, trampling by animals or people) allows urushiol to reach the surface, oxidize, and turn black.

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MOST COMMON ALLERGENIC MEMBERS OF THE ANACARDIACEAE FAMILY AND MEMBERS OF TWO OTHER PLANT FAMILIES THAT ALSO CONTAIN URUSHIOL CROSS-REACTING CHEMICALS		
Binomial name	**Common name**	**Notes**
Toxicodendron radicans	Common or Eastern poison ivy	Eastern half of USA Climbs trees with aerial rootlets Some subspecies in Mexico, Japan, China
Toxicodendron rydbergii	Northern or Western poison ivy	Western half of USA (except CA) and northern border states Non-climbing shrubs
Smodingium argutum	African poison ivy	South Africa
Toxicodendron pubescens	Eastern poison oak	Southeast USA
Toxicodendron diversilobum	Western poison oak	Pacific Coast of USA Usually a shrub but can climb
Toxicodendron vernix	Poison sumac	Eastern third of USA Not a weed Grows in standing water
Anacardium occidentale	Cashew nut tree	Tropical Americas Highest concentration of allergen in oil in middle layer of nut shell
Mangifera indica	Mango tree	Tropics (see text)
Metopium toxiferum	Poisonwood tree	Caribbean basin
Semecarpus anacardium	Indian marking nut tree	Southeast Asia, Pacific Islands, Australia Used to mark laundry in India ("dhobie mark dermatitis") and used topically as herbal remedy for dermatitis, tattoo removal
Toxicodendron verniciflua	Japanese lacquer tree Chinese lacquer tree	Japan and China Resin used to lacquer wood products
Gluta renghas	Rengas tree	Malaysia lumber source
Schinus terebinthifolius	Brazilian pepper tree or Florida holly	Native to South America, but now a problem in southern Florida, Hawaii, Central America, the Pacific Rim and Pacific Islands
Cross reactors		
Ginkgo biloba (Ginkgoaceae)	Ginkgo tree	Cultivated worldwide in temperate climates – only the seed coat is allergenic
Grevillea spp. (Proteaceae)	Spider flower, silk oak, Hawaiian tree, Kahili tree	Widely cultivated Australian natives

Table 17.10 Most common allergenic members of the Anacardiaceae family and members of two other plant families that also contain urushiol cross-reacting chemicals.

cross-reactions to other Anacardiaceae, whereas early *oral* exposure to resorcinols may induce a state of tolerance.

The 15- to 20-meter tall Japanese lacquer tree (*Toxicodendron verniciflua;* also referred to as the Chinese lacquer tree) produces a thick, self-melanizing, viscous bark sap used for varnishing wood. Because polymerized urushiol remains in the lacquer, it maintains its allergenicity for many years. Patients allergic to poison ivy usually react to Japanese lacquer tree catechols. In a survey of 232 lacquer craftsmen, 189 (81%) developed dermatitis from the lacquer, but 83% of these reactions resolved with continued lacquer exposure (natural hyposensitization)[17].

Other common urushiol-containing Anacardiaceae family members include *Smodingium argutum* (African poison ivy) in South Africa, *Toxicodendron striatum* (Manzanillo tree) in Colombia, and *Lithraea caustica* (Litre tree) in central Chile.

Cross-reactors from other families

The seed of the Ginkgo tree (*Ginkgo biloba*), the only living member of the family Ginkgoaceae, possesses the allergenic catechol ginkgolic acid. Because of the tree's large size, beauty, and resistance to air pollution, male ginkgo trees are widely grown in suburban North America and Europe. The yellowish seeds of female trees disintegrate and release rancid-smelling butyric acid. Contact with the soft outer layer (pulp) of the seed causes most allergic reactions. Dermatitis following contact with the seed kernel, seed shell, and leaves has been reported, but these parts likely only cause irritant reactions (Fig. 17.9). Handling the intact seed (without the seed pulp) does not cause dermatitis in sensitized individuals. The unique, fan-shaped leaf makes the ginkgo easy to recognize.

The genus *Grevillea* (Proteaceae) includes 250 plant species common to Australia that contain pentadecylresorcinol. Having no side-chain double bonds, it is less allergenic than poison ivy and poison oak allergens. The flower of the Hawaiian kahili tree (*Grevillea banksii*) is a significant cause of allergic contact dermatitis in Hawaii.

Asteraceae and related families

Key features

- Asteraceae (Compositae) identification
  - Flower head with all strap-like florets (e.g. dandelion)
  - Flower head with small, central tubular florets and peripheral strap-like florets (e.g. daisy, sunflower)
  - Leaf-like bracts surround undersurfaces of flowers
- Allergens are sesquiterpene lactones (SQLs)

Asteraceae identification

The compound flower heads in this largest of all plant families (~24 000 species) are composed of clusters of tiny flowers (florets) subtended by a whorl of leaf-like bracts (Fig. 17.10). In some plants, an inner "disc" of short, tubular florets is surrounded by long, strap-like "ray" florets (see Fig. 17.10A). In some plants, all the florets are similar (see Fig. 17.10C).

Allergenic Asteraceae

The daisy family includes many troublesome weeds, ornamentals, herbaceous perennials, and vegetables (Table 17.11).

In 1956, *Parthenium hysterophorus* ("congress grass"), a native of northeastern Mexico, accidentally traveled in a consignment of US wheat to India, where it found an inviting ecologic niche. It now infests approximately two million hectares within India as well as large areas of the Americas, South Africa, Madagascar, eastern Australia, and the Pacific Islands. Unlike South American *P. hysterophorus*, the Indian plant contains high concentrations of SQLs and is likely the most common cause of Asteraceae dermatitis in the world.

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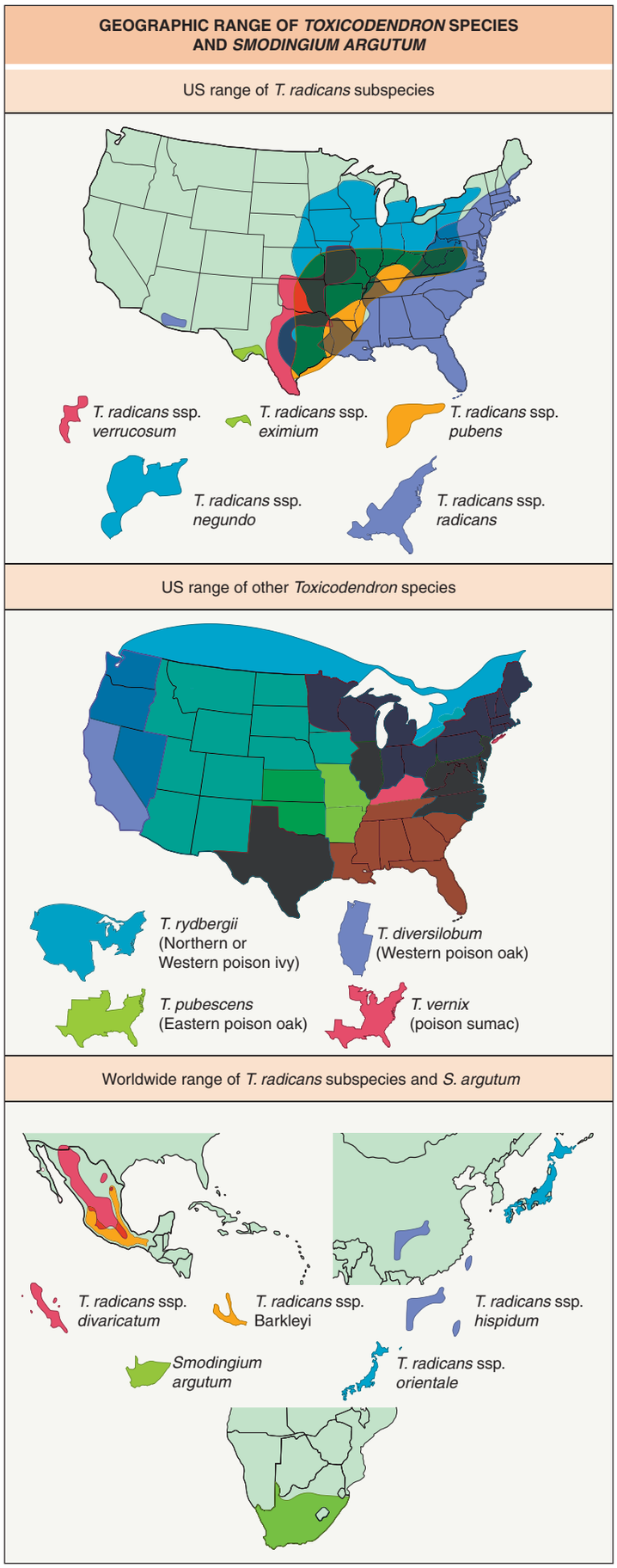

Fig. 17.8 Geographic range of *Toxicodendron* spp. and *Smodingium argutum* (African poison ivy).

Fig. 17.9 Leaves and seeds of *Ginkgo biloba*. Only the fresh seed covering contains the allergen, ginkgolic acid, which typically cross-reacts with urushiol.

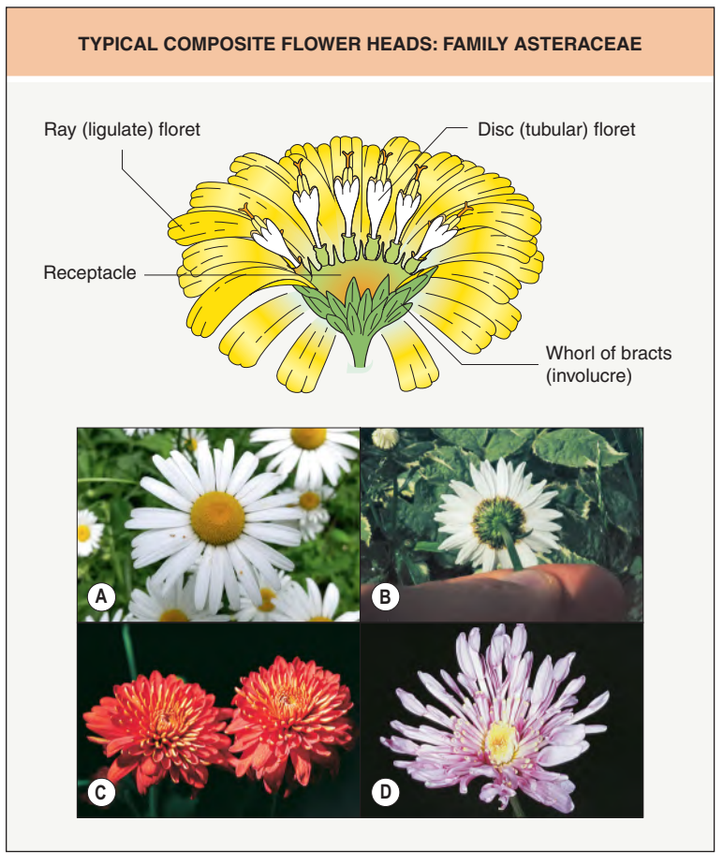

Fig. 17.10 Typical composite flower heads of members of the family Asteraceae (Compositae). Daisy (*Leucanthemum* spp.), demonstrating a composite flower head of numerous, tiny, yellow tubular florets centrally and white ray florets peripherally **(A)** as well as green leaf-like bracts beneath the flower head **(B)**, both of which are characteristic of flower heads of the family Asteraceae. **C** A cushion chrysanthemum (X *Dendranthema* cvs), demonstrating all ray, or strap-like, florets. **D** A composite chrysanthemum (X *Dendranthema* cvs), demonstrating central yellow tubular florets and peripheral ray florets.

Cross-reactors from other families

Allergic reactions to SQLs occur following contact with tulip poplar bark (*Liriodendron tulipifera*) and *Magnolia* trees (Magnoliaceae); liverworts (*Frullania* spp. [Jubulaceae]) in the Pacific Northwest; and leaves of the bay laurel tree (*Laurus nobilis* [Lauraceae]), a Mediterranean basin native.

Pathogenesis

Anacardiaceae allergens

The allergenic substance urushiol derives its name from the Japanese word for the sap (kiurushi) of the Japanese lacquer tree (*T. verniciflua*). It contains a mixture of catechols (1,2-dihydroxybenzenes) and resorcinols (1,3-dihydroxybenzenes) that avidly binds to skin[18].

Catechols and their alkyl side chains are immunologically inert. However, combining them produces potent sensitizers[19]. The

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SOME OF THE MORE THAN 200 ALLERGENIC MEMBERS OF ASTERACEAE (COMPOSITAE) FOUND THROUGHOUT TEMPERATE WORLD CLIMATES			
Category	**Binomial name**	**Common name**	**Notes**
Wild flowers and weeds	*Achillea millefolium*	Yarrow, milfoil	A cause of allergic contact "strimmer" or "weed whacker" dermatitis
	Ambrosia spp.	American ragweeds	Allergic contact dermatitis throughout growing season; allergic rhinitis only during pollen season
	Arnica montana	Arnica, mountain tobacco	
	Artemisia vulgaris	Mugwort	Grows in temperate climates worldwide
	Helenium autumnale	Sneezeweed	
	Iva spp.	Marsh elder, sumpweed	
	Parthenium hysterophorus	Scourge of India, congress grass, wild feverfew	Probably most common cause of Asteraceae dermatitis worldwide; Australian "bush dermatitis"; parthenin is primary SQL
	Tanacetum cinerarilifolium	Pyrethrum	
	Tanacetum vulgare	Tansy	
	Tanacetum parthenium	Feverfew	Perhaps the best single plant to screen for SQL allergy
	Taraxacum officinale	Dandelion	Airborne contact dermatitis after mowing
Ornamental flowers	*Dahlia* spp.	Dahlia	
	X *Dendranthema* cultivars	Chrysanthemum	Common cause of SQL dermatitis; florists remove dead flowers ("dead-head") to encourage more blooms
	Helianthus annuus	Sunflower	Windblown trichomes can cause airborne contact dermatitis
	Rudbeckia hirta	Black-eyed Susan	
In herbal medicine*	*Calendula officinalis*	Pot marigold	
	Chamaemelum nobile	Sweet/Roman chamomile	
	Inula helenium	Elecampane	
Vegetables	*Cichorium endiva*	Endive	
	Cichorium intybus	Chicory	
	Cynara scolymus	Globe artichoke	
	Lactuca sativa	Lettuce	

*Also *Achillea millefolium*, *Arnica montana*, *Helianthus annuus*, *Tanacetum parthenium*, *Tanacetum vulgare*, *Taraxacum officinale*.

Table 17.11 Some of the more than 200 allergenic members of Asteraceae (Compositae) found throughout temperate world climates. SQL, sesquiterpene lactone.

immunologic activity of alkyl side chains results from weak van der Waals forces. Longer side chains increase irritancy and allergenicity; location of the catechol ring at position 3 increases antigenicity, but placement at position 6 induces tolerance. Although virtually all individuals who are allergic to poison ivy react to the C15 catechol with two double bonds (a diolefin), only 35% respond to the C15 catechol with an unsaturated side chain (pentadecylcatechol). Therefore, pentadecylcatechol, which is easily synthesized, is an inappropriate patch test allergen for Anacardiaceae sensitivity, because the diolefin is the major component in poison ivy and poison oak urushiols.

Asteraceae allergens

The major sensitizers, sesquiterpene lactones (SQLs), are found in the leaves, stems, and flowers, and they are composed of a sesquiterpene ($C_{15}H_{24}$) and a lactone ring (cyclic ester). Direct plant contact, botanical extracts in skin care products and cosmetics, and dry airborne plant parts including pollen can transfer SQLs to individuals. Over 5000 SQLs have been described, of which about 4000 are potentially allergenic. The composition of SQLs produced by any species varies according to location and weather.

Patch testing for SQL allergy has traditionally relied on the SQL mix consisting of three SQLs – costunolide, alantolactone, and dehydrocostus lactone – or the "Compositae mix" of non-standardized extracts from five species of Asteraceae. The SQL mix has a false-negative rate of 35%–65% in SQL-allergic patients, while the Compositae mix has a false-negative rate of at least 15%. The SQLs present in these mixtures poorly represent SQL structural diversity[20]. A combination of three commercially available patch tests – SQL mix, Compositae II mix, and parthenolide (also contained in Compositae II mix) – can be used as this combination detects 96% of known SQL allergic individuals[21]. Ideally, patients should be tested to portions of the suspect plant.

Clinical Features

Anacardiaceae dermatitis

Key features

- Urushiol is a partially water-soluble allergen that must be washed off quickly
- Treat for at least 2 weeks, otherwise rebound dermatitis commonly occurs
- Eruption "progresses" to "new areas" because of variability in antigen concentration and stratum corneum thickness

Lightly brushing against uninjured leaves is innocuous, since plants must be damaged to release urushiol (see Fig. 17.7E). However, in the late fall, plants spontaneously release urushiol. In addition, contaminated clothing, pets, lacquered furniture, sawdust, and smoke (potentially causing severe respiratory tract inflammation, dermatitis, and even temporary blindness) can transfer urushiol.

After urushiol contact, a sensitized person typically develops an erythematous, pruritic eruption within 2 days (range, 4–96 hours) that peaks within 1–14 days[22] . However, dermatitis may occur up to 3 weeks after primary contact or within hours of secondary contact. Streaks of erythema and edematous papules typically precede vesicles and bullae (Fig. 17.11A,B,E). If the antigen load is lower, only erythematous, edematous reactions may be seen (Fig. 17.11C,D). Although allergic contact dermatitis is the most common cause of such a streaky, vesicular dermatitis, plants may cause this picture by other means (e.g. chemical irritant dermatitis or the initial phase of phytophotodermatitis). Patch tests of bulla fluid are routinely negative. The illusion that

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this fluid propagates the eruption results from variations in the time it takes for a clinical reaction to occur at different sites that have received differing antigen loads and have variable stratum corneum thicknesses.

Erythema multiforme appears to be an underreported sequela of severe poison ivy reactions that may appear 2 weeks following onset of the dermatitis (Table 17.12)[23]. Uncommonly, nephritis or eruptions resembling measles, scarlatina, or urticaria develop and are ascribed to immune complex deposition. Without treatment, poison ivy dermatitis lasts about 2–3 weeks. Prolonged postinflammatory hyperpigmentation may occur in individuals with darker skin phototypes.

Over 70% of the US population react to poison ivy allergens after patch testing, but only 50% react to plants in nature. Interestingly, only 15% of atopic persons are sensitive, and studies suggest that sensitivity is hereditary. Urushiol-sensitive AIDS patients with $CD4^+$ counts <200/mcl maintain sensitivity to poison ivy allergens[24] because the predominant effector T cells against poison ivy catechols are $CD8^+$.

In "black-spot" poison ivy dermatitis, urushiol acts as both irritant and allergen; an irritant contact dermatitis is superimposed on an allergic contact dermatitis[25,25a]. Oxidized resin creates a black discoloration on the skin (Fig. 17.11F). If urushiol stains clothing black, it cannot be washed off and will retain its allergenicity indefinitely. A more widespread spotted pattern in non-covered sites is seen if a patient has been weed-whacking areas of poison ivy (Fig. 17.11G).

Asteraceae dermatitis

Key features

- Caused by sesquiterpene lactones (SQLs)
- Most common cause for home gardeners is chrysanthemums
- Most common worldwide cause is *Parthenium hysterophorus*
- Often presents with an airborne contact pattern
- Patients frequently become photosensitive

Chrysanthemums probably cause more allergic contact dermatitis than any other home garden plant (Fig. 17.12). The flowers and leaves are more potent sensitizers than the stems, and contact is common when removing dead flowers to encourage further blooming ("dead-heading").

Middle-aged men with a history of outdoor exposure may develop Asteraceae allergy that resembles airborne contact dermatitis (ABCD) (Fig. 17.13). SQLs, though not volatile, are bound to airborne trichomes and dried leaves in sufficient concentrations to cause ABCD[26]. Typically, a single region of the body is involved for several years, flaring in the summer during the plant's growing season and disappearing during the winter. Later, all exposed areas develop a chronic pruritic and lichenified dermatitis. However, the eyelids, melolabial folds, retroauricular sulci and antecubital fossae are generally involved, unlike the situation in photosensitive dermatitis (see Fig. 87.15).

Ragweed (*Ambrosia artemisiifolia*) commonly causes ocular and airway immediate hypersensitivity reactions via pollen allergens, with the major allergen identified as Amb a 1. Although pollen can rapidly penetrate the skin[27], it contains no SQLs and has not been demonstrated to cause contact dermatitis. SQLs found in other parts of ragweed plants can induce ABCD (see above) and dyshidrotic eczema[28].

Parthenium hysterophorus allergic contact dermatitis accounts for 40% of all visits to contact dermatitis clinics in India and most commonly presents with an ABCD pattern (45%); 15% of patients have erythroderma while 10% develop chronic actinic dermatitis[29]. The remainder display a mixture of these three patterns. *Parthenium*-allergic patients sometimes demonstrate positive prick tests suggesting the

PLANTS AND PLANT PRODUCTS THAT CAN LEAD TO ERYTHEMA MULTIFORME IN THE SETTING OF SEVERE ALLERGIC CONTACT DERMATITIS		
Common name (binomial name)	**Family**	**Notes**
Poison ivy, oak, sumac (*Toxicodendron* spp.)	Anacardiaceae	US and Canada
Tea tree oil (*Melaleuca alternifolia*)	Myrtaceae	Australia, exported products
Wild feverfew (*Parthenium hysterophorus*)	Asteraceae	India

Table 17.12 Plants and plant products that can lead to erythema multiforme in the setting of severe allergic contact dermatitis.

Fig. 17.11 Clinical manifestations of Anacardiaceae dermatitis. A Poison ivy (*Toxicodendron radicans*) dermatitis: acute streaks of erythema with superimposed vesicles and eyelid edema with serous crusting. **B** Poison ivy dermatitis: multiple vesicles and a few hemorrhagic crusts with superimposed zinc oxide from application of calamine lotion. **C** Widespread erythema and edema associated with intense pruritus after carrying logs of the poisonwood tree (*Metopium toxiferum*) of the family Anacardiaceae. **D** Pink edematous streaks on the face. **E** Compared to **B**, more obvious erythema in association with edema and multiloculated vesicles. **F** "Black-spot" poison ivy dermatitis: note the black discoloration in the central portion of the edematous plaques due to polymerized plant oleoresin. **G** Weed whacker dermatitis with widespread spotted pattern. *A, G, Courtesy Louis A. Fragola, Jr, MD; B, Courtesy Julia Pettersen Neckman, MD; D, Courtesy Julie V. Schaffer, MD; E, F, Courtesy Kalman Watsky, MD.*

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HOUSE AND GARDEN PLANTS THAT COMMONLY CAUSE DERMATOSES

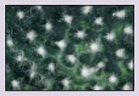
More common	Less common
X *Dendranthema* spp. e.g. cushion chrysanthemum **ACD** of hands due to "dead-heading"	*Codiaeum variegatum* i.e. florist's "croton" **Chemical irritant dermatitis**
Dieffenbachia picta i.e. dumb cane **Chemical irritant dermatitis**	*Eurphorbia pulcherrima* i.e. poinsettia **Chemical irritant dermatitis**
Primula obconica i.e. primulas **ACD** of hands due to "dead-heading"	*Calendula officinalis* i.e. calendulas (one of the "other Asteraceae") **ACD**
	Ficus benjamina variegata i.e. weeping fig **ACD, chemical irritant dermatitis**
	Pelargonium spp. e.g. geranium **ACD**
	 *Opuntia microdasys* i.e. bunny ears cactus, polka dot cactus **Mechanical irritant dermatitis** due to glochids
	Philodendron **ACD, chemical irritant dermatitis**

Fig. 17.12 House and garden plants that commonly cause dermatoses. ACD, allergic contact dermatitis. *Photograph of poinsettia courtesy Denver Botanical Gardens.*

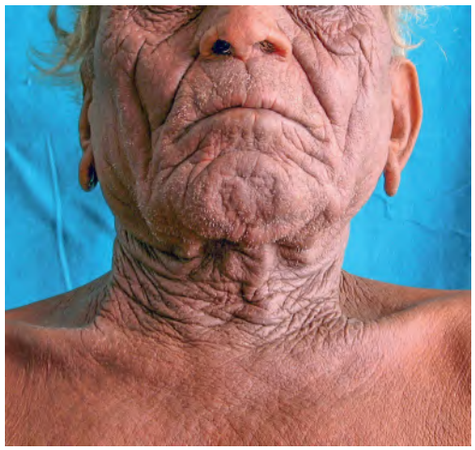


Fig. 17.13 Airborne contact dermatitis. This farmer from Gujarat, India was allergic to the sesquiterpene lactones in *Parthenium hysterophorus*. *Courtesy Shyam Verma, MBBS, DVD, FRCP.*

coexistence of type I and type IV hypersensitivity reactions. Allergens found in Asteraceae plants have been shown to cause systemic allergic dermatitis which can involve both skin and mucosae[30]. Although patients are initially sensitized via skin contact, subsequent exposure to the allergen by other routes (e.g. intravenous, intramuscular, subcutaneous, inhalation, oral, mucosal) results in dermatitis in the absence of contact of the allergen with the skin.

Photosensitivity and Asteraceae allergy

SQLs possess neither phototoxic nor photoallergic properties, yet allergic patients often develop abnormally low minimal erythema and minimal phototoxic doses to wavelengths of 300–350 nm, and conversely, up to 85% of patients with chronic photosensitivity dermatoses react to Asteraceae allergens[31]. Chronic actinic dermatitis (CAD) that develops in some patients who are allergic to SQLs is thought to be an autoimmune photodermatosis mediated by sunlight-induced antigens in the skin (see Ch. 87). For example, the SQL isoalantolactone has been shown to be highly reactive with DNA in solution during UVA irradiation, and the UV action spectrum for CAD corresponds with the UV absorption peak for DNA. An interaction between SQLs and DNA in the presence of UV light may alter DNA to render it antigenic and trigger a CAD response[32].

Although α-terthienyl and various polyacetylenes with phototoxic properties have been found in some Asteraceae, no link has been established between their presence and dermatitis[32].

Treatment (Anacardiaceae and Asteraceae Dermatitis)

The primary goal of treatment is to relieve pruritus[22]. As soon as exposure to poison ivy or related plants is recognized, one approach is to thoroughly wash the entire body three times with liquid dishwashing soap and a damp washcloth – stroked in one direction, not back and forth – under very warm to hot running water[33]. If no dishwashing soap is available, washing with just water will still remove some resin.

Weepy lesions are best treated with tepid baths, wet-to-dry soaks, or bland shake lotions (calamine) that can dry the lesions. An astringent such as Burow's solution (aluminum acetate) works well to cool and dry lesions when applied as a wet-to-dry dressing or open wet dressing. Topical antihistamines, anesthetics containing benzocaine, and antibiotics should be avoided to prevent sensitization[34,34a].

Superpotent topical corticosteroids only help if applied during the earliest stages of the outbreak when vesicles and bullae are not yet present, and their abrupt discontinuation can lead to rebound inflammation. Topical calcineurin inhibitors are ineffective in treating urushiol-induced dermatitis. One surfactant product (Zanfel® wash) reportedly relieves redness and itching[35].

Systemic corticosteroids are extremely effective when indicated. Oral prednisone should be dosed at 1–2 mg/kg/day based upon ideal weight and gradually tapered over 2–3 weeks[34]. Six-day tapering courses of methylprednisolone (e.g. Medrol® Dosepak) are too short and too low dose to be effective; rebound commonly follows completion. The author routinely uses intramuscular long-acting triamcinolone acetonide (1 mg/kg) mixed with fast-acting betamethasone (0.1 mg/kg). Patients experience rapid relief and fewer side effects than with oral prednisone[34]. Such systemic treatment is also less expensive than application of high potency topical corticosteroids[15]. Soporific oral antihistamines often induce poor-quality sleep while doing nothing for the itch, since histamine is not responsible for the pruritus.

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Hyposensitization programs for Anacardiaceae- and Asteraceae-allergic individuals have generally failed. Patients develop pruritus ani, generalized pruritus, or urticaria and comment that the "treatment is worse than the disease". It appears that hyposensitization after becoming cutaneously sensitized to the potent allergens in poison ivy and poison oak is difficult, but not impossible, as demonstrated by Japanese lacquer craftsmen (see above) and one study of *Parthenium*-allergic individuals in which 70% of patients who underwent oral hyposensitization improved symptomatically[29].

A recent study in guinea pigs suggests that vaccination may be possible. Intramuscular administration of water-soluble derivatives of certain urushiol components not only desensitized previously sensitized animals, but also prevented sensitization of naive animals[36].

Applying 5% quaternium-18 bentonite lotion before touching poison ivy reduces allergic reactions, but currently the once commercially available product is no longer available. Of note, toxicodendron allergens pass through rubber gloves, but not heavy-duty vinyl gloves.

In chronic actinic dermatitis or other widespread Asteraceae-induced dermatitis, potent topical corticosteroids and oral prednisone are relatively ineffective unless employed early (before year-round dermatitis occurs). Very-low-dose PUVA with initial oral corticosteroid coverage[37], azathioprine, or mycophenolate mofetil may be necessary to control the disease (see Ch. 87 for details).

Other Plant Families Causing Allergic Contact Dermatitis

Alliaceae

The genus *Allium* includes onions, garlic, and chives. Garlic is the most frequent cause of fingertip dermatitis in cooks. It typically presents on the thumb, index, and middle fingertips of the non-dominant hand with hyperkeratosis, desquamation, and fissuring. Patch testing to whole garlic should never be done because of its irritant properties. The most important irritant and allergen is diallyl disulfide. Onion (*Allium cepa*) dermatitis is rare. Common disposable gloves, such as vinyl, latex, polyethylene, and nitrile, do not protect sensitized individuals who handle garlic or onions. Reusable domestic rubber gloves are protective.

Alstroemeriaceae and Liliaceae

"Tulip fingers" is a combined allergic and irritant contact dermatitis caused by tulip bulbs (*Tulipa* spp.). Erythematous, scaly plaques appear on the fingertips and periungual skin, particularly on the first and second fingers of the dominant hand, but a diffuse, xerotic hand dermatitis often develops after prolonged exposure. Pulpitis, paronychia, and dissemination to the face, neck, arms, and anogenital area may ensue. Tuliposide A, a glycoside, is found in the white epidermis of bulbs. Acidic hydrolysis converts it to tulipalin A, the allergen. Tuliposide B is also found in tulips, but its hydrolytic product, tulipalin B, is a much weaker sensitizer.

Since their popularization in Holland in 1963, Peruvian lilies (*Alstroemeria aurantiaca* and *A. ligtu*) have become incredibly popular in floral arrangements. Handlers remove individual flowers, thread wire through the stems, and/or cut off leaves (Fig. 17.14A). These workers can develop erythema, fissuring, vesicles, hyperkeratosis, and exfoliation of the fingertips, greater on the dominant hand (Fig. 17.14B)[38]. Most often, allergic reactions present with non-inflamed distal finger fissuring and hyperkeratosis. Unfortunately, the allergen passes through vinyl (polyvinyl chloride) gloves[38]. Although nitrile (synthetic rubber) gloves are protective, few florists will wear them. Tuliposide A and B are found in virtually all portions of the plant. The flowers have more allergen than the stems, and the leaves have the least amount.

Myrtaceae

Since the early 1990s, oil distilled from the leaves of *Melaleuca alternifolia* (the Australian tea tree) has been increasingly reported as a cause of allergic contact dermatitis. At least two-thirds of Australians have been exposed to it, and between 2% and 7% of them are allergic to it. While tea tree oil possesses broad-spectrum antimicrobial, anti-inflammatory, and antihistaminic effects, at least 16 different constituents are strong sensitizers. Because most sensitizers are degradation products formed after exposure of tea tree oil to light, warmth, and moisture, fresh tea tree oil is a weak sensitizer[39].

Botanical products

Hundreds of botanical extracts are added to skin care products for fragrance and purported healing properties. Fifty to 60% of patients with a presumed contact allergy to botanical extracts display at least one relevant positive reaction[3,40]. In a 2004 report, the most common culprit was tea tree oil, followed by feverfew (Asteraceae family) and lichen acid mix (found in deodorants)[3]. A more recent analysis of allergic contact dermatitis due to the use of herbal remedies found that Asteraceae plants were the most common source of botanical allergens[40].

Balsam of Peru (*Myroxylon* spp.), fragrance mix, Compositae mix, and SQL mix are poor screens for botanical allergy in that they uncover only 33%, 30%, 20%, and 7% of relevant allergic reactions, respectively[3]. A *repeat open application test* performed by applying the suspected allergenic product to the antecubital fossa twice daily for seven days will help diagnose an allergy to botanical extracts.

Table 17.13 lists plant dermatoses most commonly observed by dermatologists in various countries from around the world.

OCCUPATIONAL PLANT DERMATOSES

Bakers

Bakers and food processors have high rates of immunologic contact urticaria[41]. Flours, grains, and feeds ranked third behind cow dander and latex as the causes of such urticaria[2]. Omalizumab has allowed at least one baker to maintain his job when protein contact dermatitis to wheat threatened his livelihood[42].

Florists and Horticulturalists[43–45]

Irritant dermatitis outweighs allergic contact dermatitis in frequency and importance. Estimates of hand dermatitis in florists range from an 8% point prevalence to a 25%–30% annual incidence to a nearly 50% lifetime incidence, especially amongst floral designers. Although most cases of hand dermatitis are mild, periodic, and self-limited, because florists' hands are often wet, primary irritant reactions may lead to sensitization over long periods of time.

For florists, the most common sensitizers are sesquiterpene lactones, tulipalin A, and primin (*Primula obconica*). Daffodil itch (caused by calcium oxalate), from handling stems and bulbs, is probably the most common cause of irritant contact dermatitis in florists. In one series,

Fig. 17.14 Allergic contact dermatitis in a florist. **A** Florist with fingertip *Alstroemeria* allergic contact dermatitis showing how he holds a Peruvian lily while cutting off the leaves. **B** Index finger of the same florist, showing the chronic, lichenified, hyperkeratotic dermatitis caused by allergy to Peruvian lily and tulipalin A. *Photos with permission from the American Journal of Contact Dermatitis 1999;10:172–6.*

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10% of 250 horticultural workers were allergic to Asteraceae[45] and one-third developed urticaria. An atopic family history increased one's likelihood of developing urticaria[45].

Food Handlers

Food handlers are at increased risk for immunologic contact urticaria and protein contact dermatitis (see above)[46]. Of those who develop protein contact dermatitis, over 60% need to change jobs. Among grocery store workers, contact with fresh celery combined with indoor tanning increases the prevalence ratio for phytophotodermatitis to >40 as compared to those with neither risk factor.

Outdoor Workers

Among outdoor workers, poison ivy and poison oak (*Toxicodendron* spp.) are the main causes of occupational contact dermatitis. *Toxicodendron* contact accounts for 10% of all lost-time injuries for the US Department of Agriculture Forest Service. Foresters, especially when fighting forest fires, have virtually no control over avoiding these plants. Up to 25% of firefighters need to leave the fire line because of severe *Toxicodendron* dermatitis.

Forestry workers in the American and Canadian Pacific Northwest may develop "woodcutter's eczema", an allergic contact dermatitis caused by airborne or rainwater-dispersed SQLs from liverworts (*Frullania* spp.) growing on trees and rocks. The dermatitis is worse in wet winter months and presents in a pseudo-photodistribution, involving the upper eyelids while sparing the submental area.

DERMATOSES ERRONEOUSLY ASCRIBED TO PLANTS

Organisms that are neither animal nor plant belong to one of the following kingdoms:

- Monera – unicellular prokaryotes (anuclear); includes bacteria and blue-green algae (cyanobacteria)
- Fungi – multicellular, heterotrophic (absorb food) eukaryotes (possess nuclei); lack cilia or flagella
- Protista – unicellular eukaryotic protozoans and algae

Reactions to these organisms can be falsely ascribed as reactions to plants.

Lichen Dermatitis[47]

Lichens are composite organisms consisting of a fungus and a photosynthetic partner, usually a green or blue-green alga. In the Pacific Northwest, the fungal component of lichens growing on trees, rocks, or soil disperses usnic acid, atranorin, and everinic acid onto foresters or other passersby via either direct contact or rainwater. This leads to allergic contact dermatitis in a pseudo-photodistribution. These three lichen acids are found in the common fragrance constituent oak moss absolute, and they can also cause photoallergic contact dermatitis. Oak moss absolute is one of eight ingredients in the fragrance mix I and is responsible for almost 40% of positive reactions to this fragrance mix.

Chlorella Photodermatitis

Ingestion of *Chlorella*, a nutritional supplement grown in Taiwan and Japan, can cause photodermatitis. *Chlorella* is a single-celled green alga that produces pheophorbide A, a photosensitizing breakdown product of chlorophyll that has been used experimentally for photodynamic therapy.

Seaweed Dermatitis[48]

While there are over 3000 species of seaweed, the unicellular blue-green alga *Lyngbya majuscula* is the primary species that causes a chemical irritant contact dermatitis (due to aplysiatoxin and debromoaplysiatoxin). Usually growing in fine blackish-green to olive-green strands up to 30 cm long, it looks like matted hair or felt and can get caught under bathing or wet suits. Within minutes of contact, genitalia and buttocks develop erythema, pruritus, and edema. Pustules and several days of desquamation often follow.

Seaweed dermatitis is primarily a problem during algal blooms stimulated by pollution and overfishing. In temperate and tropical oceans around the world, *Lyngbya* has been observed in intertidal zones up to a depth of 30 meters.

Pseudo-Phytodermatitis From Plant Hitchhikers

Allergic and irritant chemical reactions may occur from insecticides and other chemicals used to prevent horticultural or agricultural damage. Contaminating arthropods such as the hay itch mite (*Pyemotes* spp.) found in grains can cause pruritic wheals in dock workers, farmers,

GLOBAL PERSPECTIVE – PLANT DERMATOSES MOST COMMONLY OBSERVED BY DERMATOLOGISTS IN VARIOUS COUNTRIES			
Country	**Plants/Types of reactions**		
Australia	*Lyngbya majuscula* (seaweed dermatitis)/ICD	*Dendrocnide* tree/TMU	*Melaleuca* (tea tree oil)/ACD
Brazil	*Anacardium occidentale* (cashew nut tree)/ACD	*Mangifera indica* (mango tree)/ACD	*Schinus terebinthifolia(us)* (Brazilian peppertree)/ACD
Canada	Lichen (woodcutter's eczema)/ACD	*Frullania* (liverwort; woodcutter's eczema)/ACD	*Toxicodendron* spp. (poison ivy)/ACD
Chile	*Lithraea caustica* [Anacardiaceae]/ACD	*Ruta graveolens* (common rue)/PPD	*Urtica* spp. (stinging nettles)/TMU
China	*Toxicodendron vernicifluum*/ACD	*Mangifera indica* (mango tree)/ACD	*Ginkgo biloba* (Ginkgo tree)/ACD
Colombia	*Toxicodendron striatum* (manzanillo)/ACD	*Alstroemeria* spp./ACD	
India	*Parthenium hysterophorus*/ACD	*Chrysanthemum* spp./ACD	
Italy	*Ficus carica* (fig tree)/ICD, MID, PPD		
Japan	*Toxicodendron vernicifluum*/ACD	*Primula obconica*/ACD	
Lebanon	*Chrysanthemum* spp./ACD		
Philippines	*Lyngbya majuscula* (seaweed dermatitis)/ICD	*Anacardium occidentale* (cashew nut tree)/ACD	
South Africa	*Smodingium argutum* (African poison ivy)/ACD	*Alstroemeria spp.*/ACD	*Chrysanthemum* spp./ACD
United States	*Toxicodendron* spp. (poison ivy/oak)/ACD	*Alstroemeria* spp./ACD	*Chrysanthemum* spp./ACD
Uzbekistan	*Ficus carica* (fig tree)/ICD, MCD, PPD	*Solanum lycopersicum* (tomato leaves)/ACD, TMU	*Chelidonium majus* [Papaveraceae] (greater celandine)/ICD

Table 17.13 Global perspective – plant dermatoses most commonly observed by dermatologists in various countries. Note that *Lyngbya maluscula* is an alga. ACD, allergic contact dermatitis; ICD, irritant contact dermatitis; MID, mechanical irritant dermatitis; PPD, phytophotodermatitis; TMU, toxin-mediated urticaria.

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packers and storers of grain (see Ch. 85). Paraphenylenediamine added to henna tattoos can induce allergic contact dermatitis. Allergic contact dermatitis due to normal plant tissue levels of nickel have been reported for *Euphorbia triangularis* (Euphorbiaceae) and the freshwater plant *Ludwigia repens* (Onagraceae).

Additional figures available in our eBook (see inside front cover for access code).

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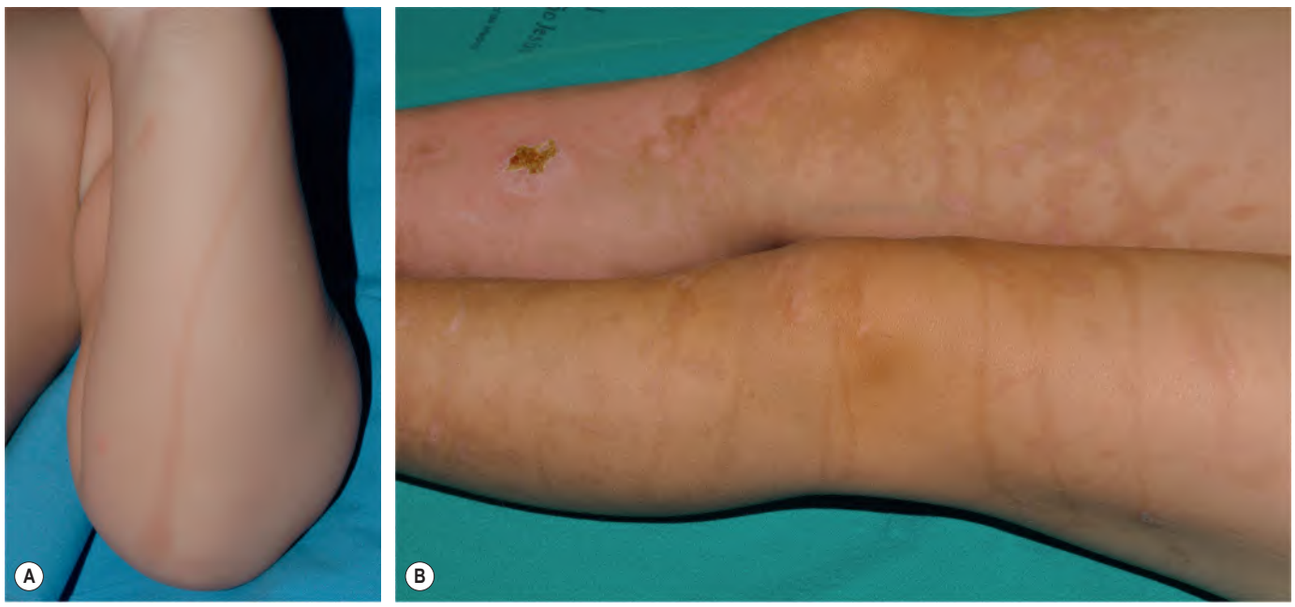
eFig. 17.1 Erythematous, burning, edematous scrotum from an irritant reaction after applying garlic (*Allium sativum*) to treat a "rash". *Courtesy Martin Giandoni, MD.*

eFig. 17.4 Phytophotodermatitis. The patient was exposed to limes then sunlight while making guacamole dip. *Courtesy Kalman Watsky, MD.*

eFig. 17.2 "Chili burns" of the non-dominant hand after cutting jalepeño peppers. Capsaicin-induced depolarization of nerves leads to vasodilation and sensory nerve activation. As a result, the patient develops erythema accompanied by a burning sensation.

eFig. 17.5 Allergic contact dermatitis due to mangos. Note the linear lesion at 5 o'clock. *Courtesy Scott Norton, MD.*



eFig. 17.3 Phytophotodermatitis. A, B Linear hyperpigmented streaks are seen in addition to inflammation and crusting in **B.** *A, Courtesy Julie V. Schaffer, MD; B, Courtesy Antonio Torrelo, MD.*

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eFig. 17.6 Poison ivy (*Toxicodendron radicans*) dermatitis. Vesicles and bullae arising within areas of edematous erythema. Pruritus can be severe. *Courtesy Kalman Watsky, MD.*

eFig. 17.7 Allergic contact dermatitis due to poison ivy (*Toxicodendron radicans*). Pink, edematous streaks on the trunk. *Courtesy Julie V. Schaffer, MD.*

eFig. 17.8 Stinging nettle (*Urtica dioica*). A Habitat on the Kansas prairie. **B** Trichomes on both the stem and the leaf surfaces. *A, Courtesy Craig A. Freeman; with permission from McGovern TW, Barkley TM. Botanical briefs. Stinging nettle – Urtica dioica L. Cutis. 1998;62:63–4; ©1998 Quadrant Health Com Inc.*

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